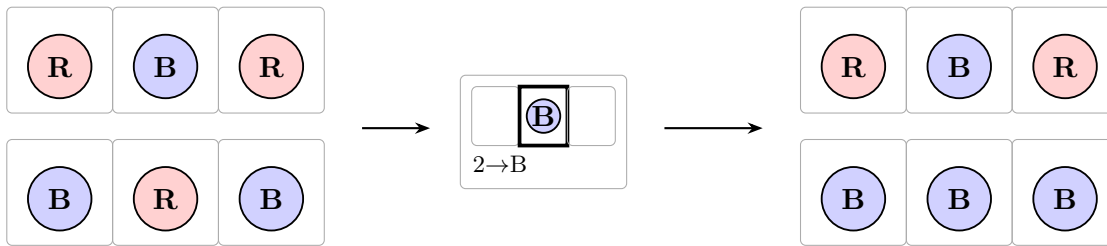
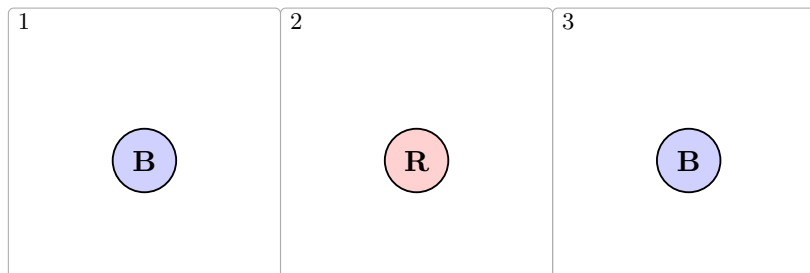
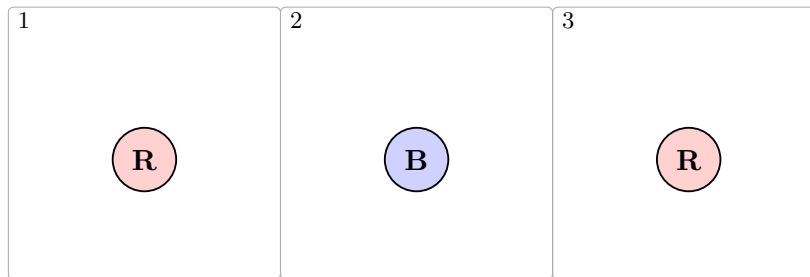


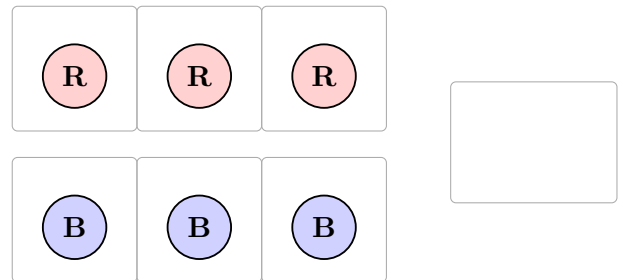
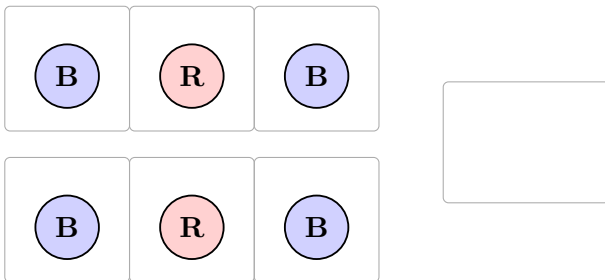
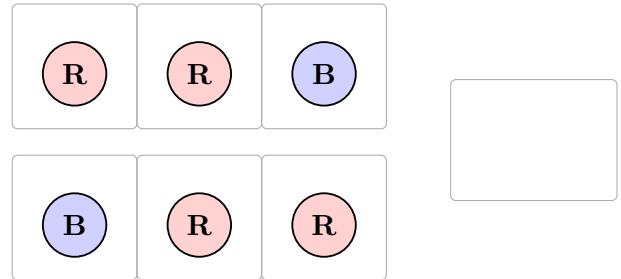
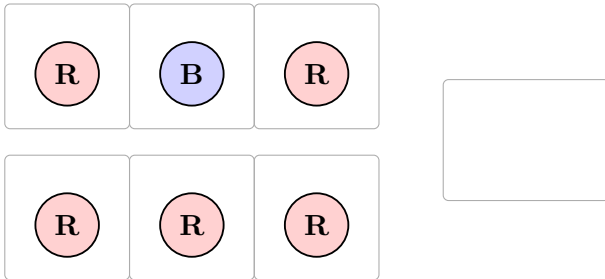
Each instruction names a slot and a color. Set that slot to that color on both boards, even if it is already that color. Slots are 1, 2, 3 from left to right.



Problem 1: Make these boards match using as few instructions as you can. Can any other starts need more instructions?



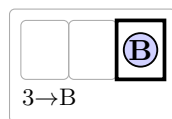
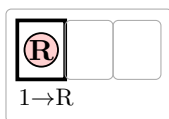
Problem 2: How few instructions can make each pair match? Try every pair.



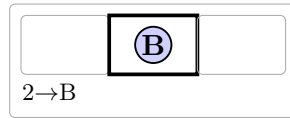
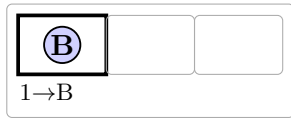
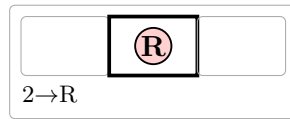
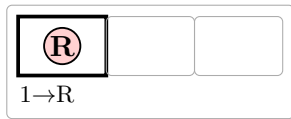
Problem 3: Invent a pair that needs exactly two instructions to match. Can you invent a pair that needs four?

Problem 4: Choose three instructions that make any two starts match. Could just two instructions always do it?

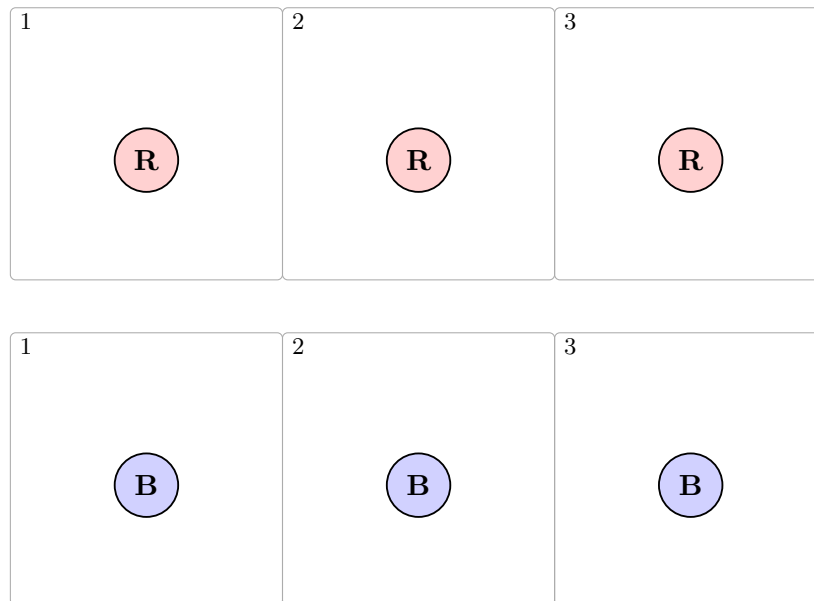
Problem 5: Use these two instructions. Make one starting pair that ends matching and another that ends different.



Mix these six cards in a cup. Draw one, use it on both boards, then put it back and mix.



Problem 6: Try six draws from these starts, then start again. Could matching take more than three draws?



Problem 7: Keep drawing after the boards match. Can any instruction make them different again?